

Accenture Innovation Challenge 2026 / Round 2

Track: PatientTriage.ai

README Document

DoorBoard

Live Emergency Department Door List:

*“Who goes next, into which open space,
on the first minutes of data.”*

Team Sap_accenture

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Demonstration only. All patients are synthetic. This is not a medical record. EHR chart write is disabled.

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1 Introduction

Generative and classical decision-support tools are often evaluated on a single patient at a time. Emergency desks do not work that way. Every few minutes the charge nurse must answer two questions together: *of everyone waiting, who goes next*, and *into which open space* (resus, bay, chair, or isolation) given live capacity and incomplete first-minute facts.

ESI (Emergency Severity Index) answers a related but narrower question well: how sick is *this one person*. It does not, by itself, rank a full waiting room against open beds. When many patients share the same ESI level (commonly ESI 3), order collapses into judgement under fatigue. DoorBoard is built for that gap.

1.A Problem Statement

Published acuity scales score one encounter in isolation. They do not produce a whole-room order matched to live open spaces. Thin vitals and missing SpO₂ must raise caution; they must not be filled with optimistic imputation. A second opinion from machine learning must not be allowed to *lower* rule-based acuity when that would create under-triage risk. Finally, any suggestion that reaches a nurse must remain a suggestion: nothing becomes chart truth until a human acts, and this build must never write back to a hospital EHR.

Without an explicit door-list product that respects those constraints, teams either (i) rely on whiteboard judgement alone under surge, or (ii) risk over-automating triage in ways that conflict with clinical accountability and with Accenture's brief that humans remain in the lead.

1.B Proposed Product and Scope

DoorBoard is a live ED door list. The engine suggests order and destination. The charge nurse confirms (**Send**), holds (**Hold**), or overrides (**Change**). Overrides carry a reason and land in an audit log. Feedback may propose weight changes for a named clinical owner; proposals are never auto-applied.

Scope of this Round 2 build:

- Synthetic cohort of 30 patients in `doorboard/data/patients.json`.
- Rule scores via NEWS2 (adults) and PEWS-style bands on APLS/PALS age strata (children).
- Lift-only calibrated risk model (may raise urgency; never lowers it).
- Floor UI for the nurse path; Demo panels for evidence and governance.
- Read-only integration mocks only; no live FHIR/HL7 write-back.

Design rule we will not break: where facts are thin, the engine escalates; it does not invent certainty.

2 Implementation Approach

We treat the **waiting room**, not the single patient, as the unit of decision. The build follows six steps. Each step exists because the previous one leaves a real operational gap.

2.A Score with published, age-appropriate tools

Adults are scored with **NEWS2** (Royal College of Physicians, [NEWS2](#)). Children are scored with **PEWS-style** bands against **APLS/PALS** age strata (infant, toddler, preschool, school-age, adolescent), not one blunt 0–17 bucket.

We also compute an **information** score: how complete the first-minute facts are. Missing vitals contribute zero clinical points; they are not imputed. A missing SpO₂ on an ESI 3+ patient means the engine refuses to mark them “can wait.”

In an earlier iteration we used a single paediatric band. A sixteen-year-old with a normal resting heart rate was flagged as concerning because infant thresholds were applied. That is the same class of error the brief warns about, one level down. We fixed it with five strata and expose the before-and-after in the Demo to Scales panel.

2.B Add a second opinion that can only raise urgency

A calibrated gradient-boosted model predicts $P(\text{critical within four hours})$. It is trained on a **synthetic** cohort whose generative process is written in code so a reviewer can inspect it (`doorboard/engine/ml.py`).

The blend rule is asymmetric on purpose: the model may **lift** acuity; it may **never** lower it (gated at $p \geq 0.45$). If the model is wrong in the unsafe direction, the rules floor still holds. If the rules under-call someone, the model can catch them.

2.C Rank the whole room against open space

DoorBoard consumes ESI (and its ties) and produces a whole-room order matched to open resus / bay / chair / isolation. Capacity is part of the decision, not a separate whiteboard (`doorboard/engine/ranking.py`).

2.D Watch the queue

Deterioration trends on serial vitals and wait ceilings by ESI level feed a bounded “Needs your eye” worklist (worse, long wait, missing facts, may leave) so the charge nurse is not hunting through thirty rows when the night is loud (`doorboard/engine/deterioration.py`).

2.E Keep the human in control

Send, **Hold**, and **Change** are explicit. Dangerous moves require safety confirmation. Overrides carry a reason and land in an audit log. A triage tool that quietly retunes itself from the busiest shift drifts without explanation; therefore weight proposals are review-only (`doorboard/engine/feedback.py`).

2.F Measure both kinds of error

Under-triage costs more than over-triage. We escalate under uncertainty, report **under-triage and over-triage together**, and expose the threshold curve we chose from (`doorboard/engine/evaluation.py`). Claims without a measurement path are not claims.

3 Solution Architecture

3.A Layered design

Table 1: Software layers and responsibilities.

Layer	Responsibility
UI (<code>app.py</code>)	Streamlit. Floor for the nurse path; Demo for evidence and governance.
Domain models	Patient, room state, hospital profile, audit events.
Clinical bands	NEWS2 / PEWS / APLS strata and listed deviations.
Text parsing	Negation-aware complaint and appearance matching.
Scoring	Acuity, information score, escalate rule.
Deterioration	Trend over serial vitals; wait ceilings.
ML	Calibrated risk model; lift-only blend; synthetic cohort.
Ranking	Whole-room order, space suggestion, short “why” line.
Feedback	Overrides to reviewable proposals (human-approved only).
Evaluation	Confusion matrix, triage rates, threshold sweep, wait replay, latency.
Pipeline	Load patients, surge, profiles, integration mocks, proofs.
Data	<code>doorboard/data/patients.json</code> : 30 synthetic patients.
Tests	<code>tests/test_engine.py</code> : 19 tests, no <code>pytest</code> required.

3.B Repository map

<code>app.py</code>	Streamlit UI (Floor + Demo)
<code>doorboard/engine/</code>	
<code>models.py</code>	Patient, RoomState, HospitalProfile, AuditEvent
<code>bands.py</code>	Age strata, NEWS2, PEWS/APLS
<code>text.py</code>	Negation-aware parsing
<code>scoring.py</code>	Acuity + information + escalate
<code>deterioration.py</code>	Trends and wait ceilings
<code>ml.py</code>	Calibrated GBM, lift-only blend
<code>ranking.py</code>	Order, space, why-line

feedback.py	Override proposals
evaluation.py	Rates, sweep, replay, latency
pipeline.py	Load, surge, profiles, mocks, proofs
doorboard/data/patients.json	
tests/test_engine.py	
requirements.txt	

3.C Runtime shape (nurse path)

1. Sign in (demo accounts: **Priya Nair**, charge nurse, or viewer).
2. **Floor**: Open spaces strip, then **Next up** (one person + In five seconds), then waiting list, then **Your desk**.
3. Act with **Send / Hold / Change**. Floor events (for example, patient getting worse while a bay frees) reorder the room; the nurse confirms.
4. **Busy night** applies a $3\times$ surge with a shifted acuity mix.
5. **Demo**: Evidence, Model, Scales, Feedback, Privacy / DPDP for judges.

3.D Decision flow (architecture story)

Figure 1 is the whole product story in one pass: first-minute facts and live open spaces feed a rule score (NEWS2 / PEWS-APLS + information). A lift-only ML stage may raise urgency but never lower it. The engine ranks the room against open spaces, surfaces a suggestion (who next / which space / why), and waits for the charge nurse. Confirmation writes an audit event; EHR chart write remains off.

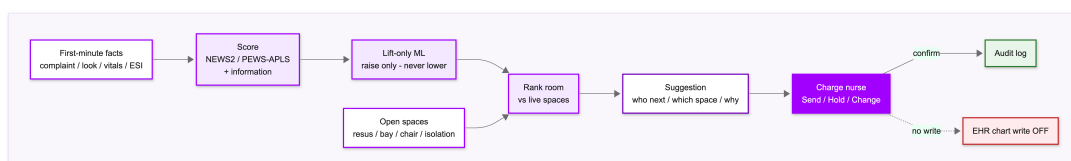


Figure 1: DoorBoard solution architecture: facts and open spaces to score, lift-only ML, room ranking, suggestion, charge-nurse confirmation, audit log, and EHR write off.

3.E Integration posture

Read-only mocks only. No live FHIR / HL7 write-back. If feeds are absent, the app degrades to the manual intake already in the prototype; it does not stop working. Three hospital profiles (urban / district / rural) change capacity and volume configuration without rewriting the scoring engine. Governance notes (DPDP ss. 7 / 6 / 9 and DPIA) live in GOVERNANCE.md in the repository package.

4 Dependencies

Python **3.11+** is recommended (3.12 / 3.13 fine). After install, **no network** and **no API keys** are required. Supported on macOS, Linux, and Windows with a modern browser.

Dependencies are pinned by minimum version in `requirements.txt`:

Table 2: Runtime dependencies.

Package	Minimum	Role
streamlit	1.32	Interactive Floor / Demo UI (docs.streamlit.io)
pandas	2.0	Tables and evidence views (pandas.pydata.org)
scikit-learn	1.4	Calibrated risk model (scikit-learn.org)
numpy	1.26	Numerics (numpy.org)

Install command:

```
python3 -m pip install -r requirements.txt
```

5 Execution Instructions

5.A Install and run locally

From the repository root (the folder that contains `app.py` and `requirements.txt`):

```
python3 -m pip install -r requirements.txt
python3 -m streamlit run app.py
```

Open <http://localhost:8501>.

First launch trains and caches the risk model (a few seconds). Later launches are faster. Streamlit configuration, if present, is under `.streamlit/config.toml`.

5.B Verify the engine

```
python3 tests/test_engine.py
```

Expect **19/19** passing. These tests re-derive the evaluation numbers in Section 6; they are not decorative.

5.C Suggested path for a judge (about 3 minutes)

1. Sign in as **Priya Nair (Charge nurse)**. Stay on **Floor**.
2. Read **Open spaces**, then **Next up** / In five seconds, then **Send**.
3. Trigger a floor event (patient getting worse / bay free), watch the list reorder, then **Accept move**.

4. Try to move a held patient down; safety confirmation blocks.
5. **Busy night** ($3 \times$ surge).
6. Switch to **Demo** then **Evidence** for under-triage / over-triage / latency.
7. Optional: **Scales** (paediatric band fix), **Privacy** / **DPDP**.

5.D Hosted demo

The live prototype is deployed on Streamlit Community Cloud:

<https://doorboard-sap.streamlit.app/>

No local install is required for judges. Sign in as **Priya Nair (Charge nurse)** and follow the path in Section 5.C. Source repository: <https://github.com/arathii23/accenture>.

6 Evaluation Evidence

Every figure below is computed by `doorboard/engine/evaluation.py` on a 30-patient reference set. Re-derive it in Demo to Evidence or via the test suite. Nothing here is typed into a slide by hand.

Table 3: Triage rates on the synthetic reference set.

	Normal shift	$3 \times$ surge
Under-triage	0.0%	0.0%
Over-triage	13.3%	23.3%
Exact tier match	86.7%	76.7%
Critical sensitivity	100%	100%

Table 4: Time to first assessment (intake stream at 3 minutes per patient).

Ordering	Mean wait, unstable	Worst unstable
DoorBoard	23.2 min	48.0 min
ESI-only	29.4 min	69.0 min
First-come-first-served	59.4 min	87.0 min

Latency is measured on local CPU with no network and excludes UI render (`evaluation.latency()` on the same 30-patient reference set). Typical values are about **0.4 ms** to score one patient and about **34 ms** median to re-rank the room of 30. Absolute wall-clock values vary slightly by machine; refresh them from Demo to Evidence or the test suite.

Honest limits. Reference labels are team consensus over synthetic patients, not clinician-signed ground truth. The ML cohort is synthetic. The rule-engine cutoff is swept on this same set. These numbers prove that the measurement apparatus exists and that the safety tuning is stated, not that DoorBoard is clinically finished for a live hospital.

7 Out of Scope

Stated exclusions for this Round 2 demonstration:

- Live hospital EHR write.
- Clinical validation on real patients.
- Diagnosis, discharge, or treatment advice.
- Replacing a clinician's judgement.

8 Conclusion

DoorBoard closes a concrete operational gap: whole-room order and destination against live open spaces, with published clinical bands, lift-only ML, human confirmation, and measurable under-triage and over-triage. The implementation is intentionally small, auditable, and offline-capable. The AI suggests; the charge nurse decides; the chart is never written by this build.